=>INTRODUCTION:

=>Significance and Purpose of the Smart-Irrigation System Project

This project automates a greenhouse using sensors and actuators to maintain optimal plant growth

conditions. Instead of manually monitoring temperature, soil moisture, and light, the system automatically

adjusts based on real-time data.

=>Key Purposes:

1. Automatic Watering – Uses a soil moisture sensor to detect dry soil and activates a sprinkler (servo

motor) to water plants.

2. Temperature Control – Uses a TMP36 sensor to monitor temperature and turns on a fan (motor) if it

gets too hot.

3. Light Control – Uses an LDR (light sensor) to detect darkness and turns on an LED to provide extra

light.

4. Real-Time Monitoring – Displays moisture, temperature, and light levels on an LCD screen for easy

monitoring.

=>Why is it important?

1.Saves water by only watering when needed.

2.Protects plants from extreme heat.

3.Improves efficiency by reducing manual work.

4.Useful for farmers and gardeners to automate plant care.

=>Overview of Components and Their Role

1. Soil Moisture Sensor – Detects soil moisture levels and helps in automatic watering.

2. Temperature Sensor (TMP36) – Measures temperature to control cooling systems.

3. Photoresistor (LDR) – Detects light intensity to control artificial lighting.

4. LCD Display – Shows real-time values of soil moisture, temperature, and light.

5. Motor (Fan) – Turns on when it gets too hot, maintaining a suitable temperature.

6. LED – Provides additional light when natural light is insufficient.

7. Servo Motor – Controls a sprinkler to water plants when the soil is dry.

8. Arduino Board – Acts as the brain, processing sensor data and controlling all components.

ARDUINO CODE:

```
#include <LiquidCrystal.h>

#include <Servo.h>

LiquidCrystal lcd(7, 6, 5, 4, 3, 2);

const int soilPin = A2;

const int tempPin = A0;

const int ldrPin = A1;

const int motorPin = 10;

const int ledPin = 13;

const int servoPin = 9;

Servo sprinkler;

void setup() {

lcd.begin(16, 2);

sprinkler.attach(servoPin);

pinMode(motorPin, OUTPUT);

pinMode(ledPin, OUTPUT);

}
```

```
void loop() {
int soilValue = analogRead(soilPin);
int moisture = map(soilValue, 0, 1023, 0, 100);
int tempValue = analogRead(tempPin);
float voltage = tempValue * 5.0 / 1023.0;
float temperature = (voltage - 0.5) * 100.0;
int lightValue = analogRead(ldrPin);
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("Moist:");
lcd.print(moisture);
lcd.print("%");
lcd.setCursor(0, 1);
lcd.print("Temp:");
lcd.print(temperature);
lcd.print("C");
delay(1000);
if (moisture < 30) {
sprinkler.write(90);
delay(2000);
sprinkler.write(0);
}
if (temperature > 30) {
digitalWrite(motorPin, HIGH);
}
else {
digitalWrite(motorPin, LOW);
```

```
}
if (lightValue < 500) {
digitalWrite(ledPin, HIGH);
} else {
digitalWrite(ledPin, LOW);
}
delay(1000);
}
```

=>Working Explanation of the Smart-Irrigation System

This project monitors and controls soil moisture, temperature, and light to create an automated greenhouse environment. The system ensures plants receive the right amount of water, ventilation, and light without human intervention.

1.Soil Moisture Monitoring & Sprinkler System:

A soil moisture sensor measures the water content in the soil.

If the moisture level drops below 30%, the servo motor activates the sprinkler to water the plants.

Once watering is done, the servo returns to its original position.

2.Temperature Control (Cooling Fan System):

A TMP36 temperature sensor monitors the greenhouse temperature.

If the temperature exceeds a set limit, the fan automatically turns ON to cool the greenhouse.

Once the temperature returns to normal, the fan turns OFF.

3. Light Control System

A photoresistor (LDR) detects ambient light levels.

If the light level is too low, the LED turns ON to provide additional lighting.

If there is enough natural light, the LED remains OFF to save energy.

4. LCD Display for Monitoring

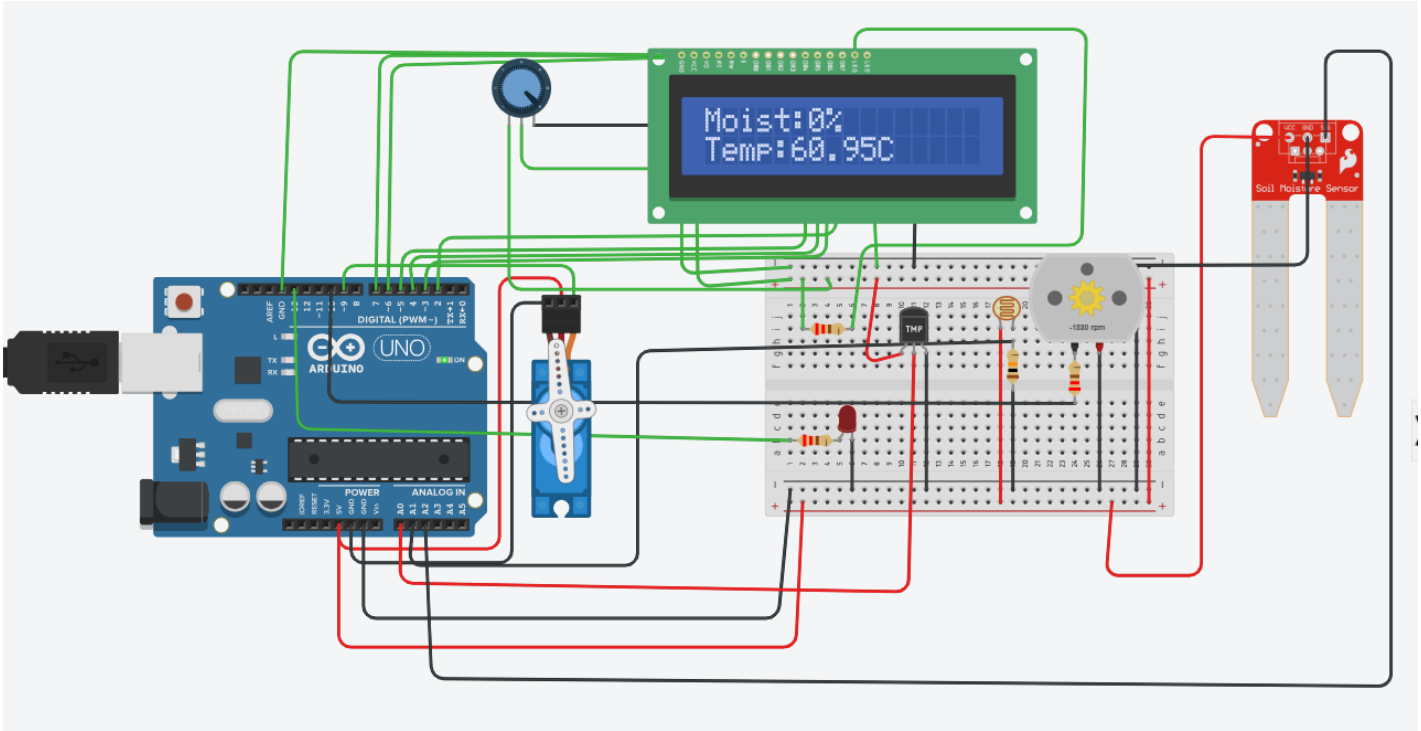
A 16x2 LCD screen continuously displays real-time values of soil moisture and temperature.

This allows users to monitor greenhouse conditions at a glance.

5. Arduino as the Control Unit

The Arduino reads all sensor data and makes decisions based on preset conditions.

It controls the servo, fan, and LED to maintain optimal conditions for plant growth.

PROJECT LINK= https://www.tinkercad.com/things/1wxqytu8Yn7-project-1

PROJECT PREVIEW: